

A Connectome Test of the Fly Hashing Algorithm

Sebastian Senge
Independent Researcher

Abstract—Dasgupta, Stevens and Navlakha (2017) showed that the *Drosophila* olfactory circuit, modelled as a random sparse projection followed by winner-take-all, is a locality-sensitive hash that beats classical LSH. The projection was random because the wiring was unknown. We test it against four electron-microscopy connectomes (MaleCNS, hemibrain, FlyWire, BANC; four animals, seven hemispheres) and report three findings. First, the 2017 pattern holds in a reimplement of its protocol on SIFT, GloVe, MNIST and odour mixtures: the fly hash beats k Gaussian projections at short hash lengths ($3.1\times$ in AP@200 on MNIST at $k = 4$). Second, that advantage is per active cell, not per operation: Gaussian projections given the same projection arithmetic retrieve better on every dataset and at every input dimension tested. Third, the connectomes behave like degree-preserving random matrices, retrieving slightly worse than their rewirings (median -1.6%; below the null on odours in all 7 hemispheres). The cost lies in the skewed fan-out of glomeruli: at equal connection count, evening out fan-out improves retrieval by up to +18%, while evening out inputs per cell does not. The random idealisation of 2017 is therefore justified, and slightly optimistic.

Index Terms—connectomics, locality-sensitive hashing, similarity search, replication

I. INTRODUCTION

Locality-sensitive hashing (LSH) maps items to short codes under which similar items collide with high probability [1], [2]. Dasgupta *et al.* [3] noticed that the fly olfactory circuit computes a related transform (Fig. 1a): about 50 glomeruli project through a sparse binary matrix onto about 2,000 Kenyon cells, and inhibition keeps the most active 5%. On SIFT, GloVe and MNIST this fly hash beat classical LSH by up to about $3\times$. The projection was random, following the account that the wiring is effectively random [4].

That account has since been revised: projection-neuron input to Kenyon cells is structured [5], and four connectomes now give the complete projection [6], [10], [11], [13]. We ask whether the random matrix was a harmless idealisation for the task that motivated it; [5] studied odour discrimination, not retrieval. Our contributions:

- a reimplement of the 2017 protocol with documented provenance, extended with odour mixtures, that recovers the published pattern;
- a budget analysis showing that the fly hash’s advantage is per active cell and not per operation;
- a comparison of the fly hash across four connectomes and seven hemispheres, on the 2017 benchmarks and on measured odour responses, against degree-preserving rewiring and against controls that change one degree sequence at a time at equal connection count.

II. RELATED WORK

The fly hash has been extended with dense pseudo-hashes and multi-probe binning [18], and with projections learned by local plasticity rules [14], [15]; these change the algorithm, whereas we keep it and change only the matrix. On the biological side, random convergence of projection-neuron input was reported from anatomical tracing [4]; electron microscopy later mapped the mushroom body completely in the larva [16] and in the adult hemibrain [17], and found structured sampling in FAFB, where simulated odour discrimination was worse than with random wiring [5]. None of these studies evaluated similarity search, the task behind the hashing interpretation.

III. METHODS

A. Connectomes and matrices

Let $M \in \{0,1\}^{d \times m}$ with $M_{ji} = 1$ iff glomerulus j connects to Kenyon cell i . For every connectome the same rule applies: uniglomerular olfactory projection neurons are assigned to glomeruli by cell type (multiglomerular, thermo- and hygro-sensory types excluded), Kenyon cells are those annotated on the same side, every reconstructed connection counts (no synapse threshold), and cells without olfactory input are dropped. Hemibrain glomerulus names are mapped to current nomenclature via the cross-dataset matches of [12]. We use MaleCNS v1.0 (male, both hemispheres), hemibrain v1.2 (female, right), FlyWire v783 and BANC v888 (female, both). All share 49 glomeruli; they have 1,761 to 2,406 Kenyon cells with 5.00 to 6.05 inputs per cell. The datasets differ in imaging, synapse detection and proofreading, so part of any difference between them is technical. The primary connectome is MaleCNS right ($d = 51$, $m = 1,886$).

Only M changes between conditions. *The null* is uniform over binary matrices with the same row and column sums as M , sampled by curveball trades [7], [8] (30 per cell; co-occurrence $Q = \sum_{j < l} (MM^T)_{jl}^2$ is stable from 1 sweep on); it keeps both degree sequences and randomises which glomeruli share a cell, and tests against it are approximate Monte Carlo randomization tests. *Equal-connection controls* have exactly $\text{nnz}(M)$ ones with inputs per cell made even, fan-out made even, or both, and random pairing. *The 2017 construction* gives each cell six uniform inputs, changing both degree sequences and the connection count.

B. Hash, scores and budgets

The fly hash computes $y = M^T x$ and sets the k largest entries of y to one. Retrieval is scored by $\text{AP}@n = \frac{1}{n} \sum_i \text{prec}@i \cdot \text{rel}(i)$ over the n true neighbours, so neighbours never retrieved count as misses, together with $\text{recall}@n$. LSH

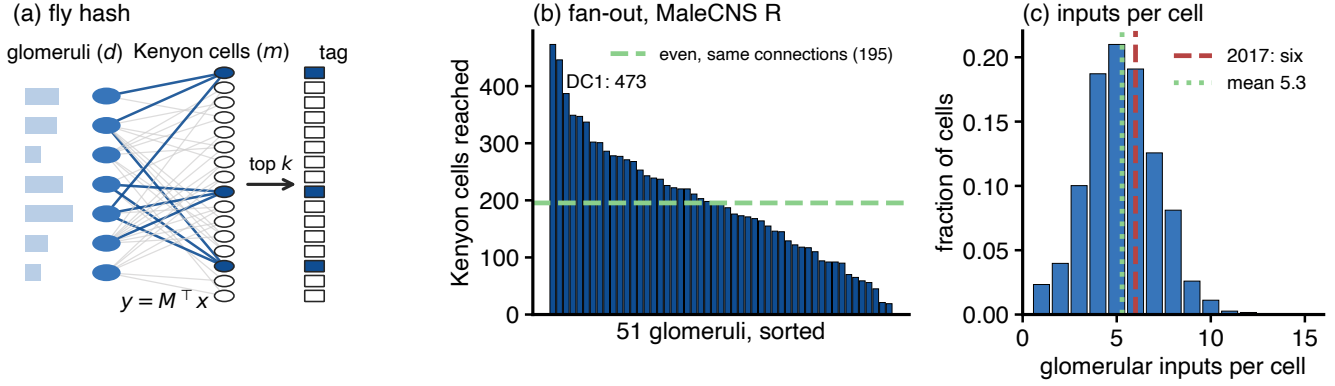


Fig. 1. (a) The fly hash: glomerular input x is expanded through a binary projection M onto Kenyon cells and the k most active cells form the tag. (b) In the MaleCNS connectome glomeruli reach very different numbers of Kenyon cells; the dashed line is the even fan-out at the same number of connections. (c) Inputs per Kenyon cell, against the six of the 2017 construction.

TABLE I
PROVENANCE OF THE PROTOCOL: 2017 MAIN TEXT [3], 2018 FLYLSH CODE [18], OR OUR CHOICE.

element	choice	source
data	10,000 vectors: SIFT-small base, 2017 (size); ours GloVe 6B 300d and MNIST train (subset) (seeded subsets)	2017 (subset)
queries, truth	1,000 queries; top 2% by Euclidean distance on raw features	2017
preprocessing	each vector centred by its own mean	2017
fly	$m = 20k$ or $10d$; top- k winners	2017
sampling	each cell samples 10% of inputs	2018 code
tag	binary; Hamming distance, random tie- break	2018 code; ours
LSH	k real-valued Gaussian projections	2017
score	AP@200 and recall@200	ours (2017: unde- fined “mAP”)
trials	50; matrices and queries redrawn	2017

computes k Gaussian projections kept as real values, as in the 2017 comparison, or their signs for bit codes. Budgets are per active cell (k projections for k winners), per stored bit, and per operation, counted as projection arithmetic as in [3]: d multiplications and d additions per projection, one addition per non-zero of M ; winner selection, retrieval and any PCA fit are excluded.

C. Approach 1: the 2017 protocol

Table I gives the protocol and the source of each element; we add 10,000 mixtures of DoOR odour profiles as a fourth dataset. The 2017 paper reports “mean average precision” without a formula. Precision averaged over *retrieved* true neighbours only, a convention in parts of the hashing literature, reproduces its LSH baseline (MNIST, $k = 4$: 0.159 against 0.160) but not its fly score (0.365 against 0.448). That score ignores misses, so we report AP@200 and recall@200 and compare with the paper through orderings and ratios. With a connectome, M accepts only its own glomeruli: odours use the glomeruli DoOR covers, and SIFT, GloVe and MNIST are reduced by PCA to 51 components assigned to glomeruli by a fresh random permutation in each trial, so these datasets

test the wiring’s statistics, not its odour-specific pairing. Each connectome is scored against 50 nulls and one draw of each control per trial.

D. Approach 2: odour analysis

Responses are the DoOR 2.0 consensus matrix [9] (commit db323a4). Coverage is uneven: 35 glomeruli are measured for 51 or more odorants, eleven for 12 to 24, five for none. We keep the 35 (our threshold of 40 odorants falls in that gap) and odorants measured on at least 12 of them: 172 odorants, 28.7% of entries unmeasured and set to zero. The benchmark is 4,000 linear mixtures of two to five profiles, normalised to unit mean as in the circuit, with the $\kappa = 10$ nearest items on raw responses as ground truth. The primary size $k = 92$ (5% of cells) and secondary sizes $\{8, 16, 32, 64\}$ were fixed in an earlier revision, after first results had been seen; this is not a preregistration. MaleCNS is tested against 200 nulls with a two-stage bootstrap (resampled odorants, regenerated mixtures, 20 nulls drawn from the same pool per replicate, 200 replicates) and a $\pm 5\%$ equivalence margin; every hemisphere is tested against 100 nulls on its own and on the 34 shared DoOR glomeruli.

IV. RESULTS

A. The 2017 pattern holds

With random matrices the published ordering is recovered on SIFT, MNIST and odours (Fig. 2): random tag selection performs like LSH, as [3] argue it must, winner-take-all at $m = 20k$ beats both, and $m = 10d$ does better still. The advantage is largest at short hash lengths ($3.1\times$ on MNIST at $k = 4$; recall $1.7\times$; $1.7\times$ on odours) and shrinks with k ; on odours and GloVe LSH overtakes the fly hash from $k \approx 8$ –16. Scoring against neighbours on centred input, as the FlyLSH code does, leaves the MNIST ratio at $3.1\times$. GloVe is close to chance at small k for every method.

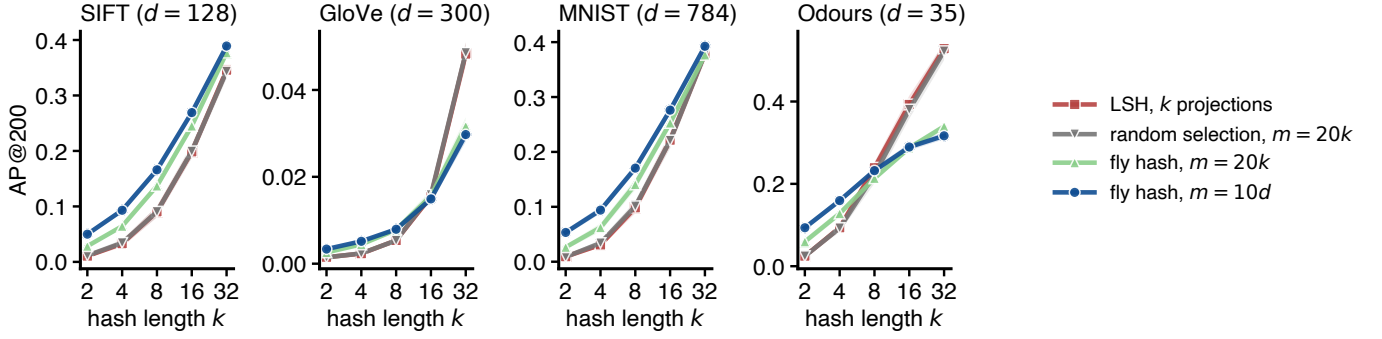


Fig. 2. The 2017 protocol with random matrices: AP@200 against hash length, mean $\pm$ SD of 50 trials. Random tag selection performs like LSH; winner-take-all at $m = 20k$ and $10d$ beats both at short hash lengths.

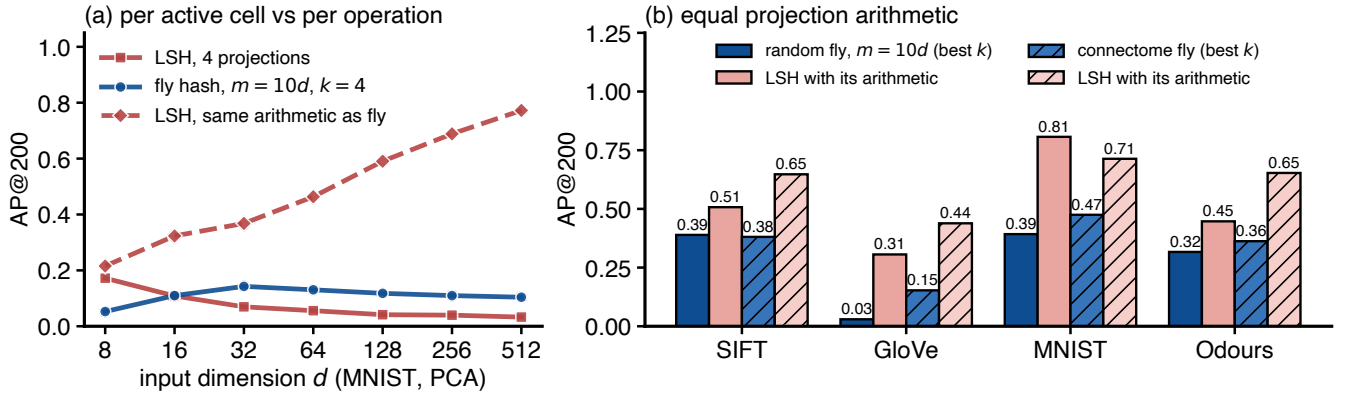


Fig. 3. (a) MNIST reduced by PCA to d dimensions, $k = 4$: the fly hash’s lead over LSH per active cell grows with d , but LSH given the same projection arithmetic as the $m = 10d$ fly leads at every d . (b) Best AP of the fly hash, with a random matrix ($m = 10d$) and with the MaleCNS connectome, against real-valued LSH given the same projection arithmetic.

TABLE II
ODOURS, MALECNS: AP OF THE FLY HASH AND OF ONE-BIT GAUSSIAN CODES (BITS IN PARENTHESES).

k	fly	sign, k bits	storage-matched	operation-matched
8	0.197	0.019	0.264 (72)	0.305 (102)
16	0.288	0.069	0.340 (130)	0.305 (102)
32	0.353	0.151	0.394 (229)	0.305 (102)
64	0.396	0.249	0.436 (397)	0.305 (102)
92	0.406	0.296	0.454 (523)	0.305 (102)

B. The advantage is per active cell, not per operation

The 2017 comparison sets k active cells against k projections, which equates projection arithmetic only at $m = 20k$. The headline network has $m = 10d$ cells sampling about $d/10$ inputs each: roughly d^2 additions against $2kd$ for LSH, a factor near $d/2k$ ($98\times$ on MNIST at $k = 4$). A sweep over input dimension on one dataset (Fig. 3a) shows both sides: per active cell, the fly hash trails LSH at $d = 8$, ties at 16 and leads from 32 on; given equal arithmetic, real-valued LSH leads at every d . The same holds on all four datasets (Fig. 3b), for random matrices (LSH 0.31–0.81 AP) and for the connectome, whose

$\text{nnz}(M)$ additions buy 98 projections that beat it on 4 of 4 datasets.

One-bit codes are weaker than real-valued projections, and against them the comparison is closer (Table II): on odours the fly hash beats sign codes of k bits ($1.37\times$ at the primary size), trails at matched storage ($0.90\times$), and at matched operations trails for small k and leads from $k = 32$ ($1.33\times$).

C. Connectomes behave like degree-preserving random matrices

Substituting a connectome for the random matrix keeps the fly hash’s advantage: it leads LSH on MNIST at $k = 4$ by 2.2 – $2.9\times$ in every hemisphere (Table III). Against its own null it is close but slightly worse (Fig. 4b): across 168 hemisphere, dataset and size combinations the median difference is -1.6% (interquartile range -3.2% to -0.3%), negative in 134, below every null draw in 59 and above every draw in 2, with a negative mean in 4 of 4 animals. Seven hemispheres come from four animals and are not independent; one male cannot separate sex from individual or reconstruction differences.

On odours (Fig. 4c) the difference at the primary size is negative in all 7 hemispheres, -3.0% to -0.6% (3 at $p < 0.05$),

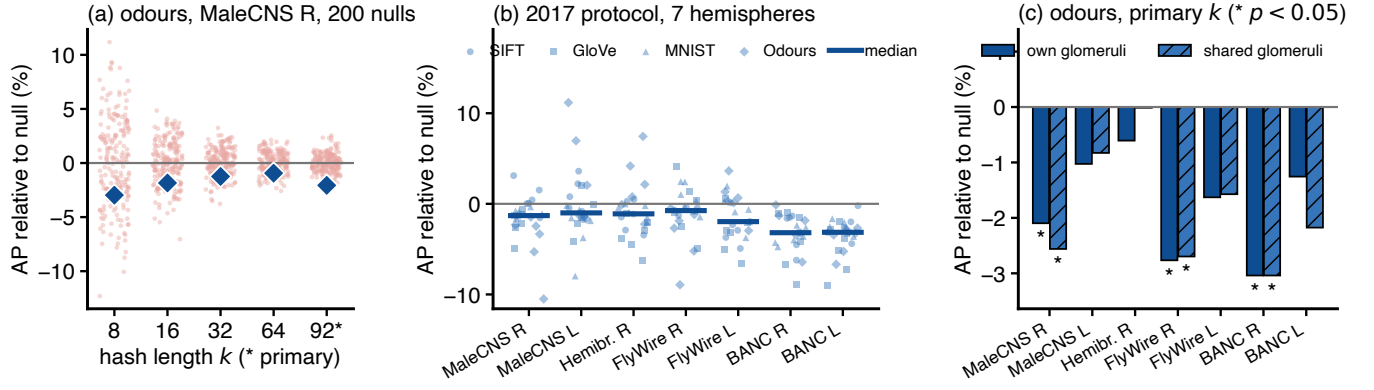


Fig. 4. Connectome against its own null (AP relative to the null mean). (a) Odour analysis, MaleCNS right: orange, 200 null matrices; blue, the connectome. (b) The 2017 protocol in seven hemispheres: four datasets $\times$ six hash lengths each, bar at the median. (c) Odour analysis at the primary size in every hemisphere, on its own DoOR glomeruli and on the 34 all share.

TABLE III

SEVEN HEMISPHERES OF FOUR ANIMALS: KENYON CELLS, MEAN INPUTS PER CELL, FAN-OUT COEFFICIENT OF VARIATION, CO-OCCURRENCE Q AS z AGAINST 100 NULLS; ODOUR ANALYSIS AT THE PRIMARY SIZE ON OWN AND ON SHARED GLOMERULI (RELATIVE DIFFERENCE %, p , FLOOR 0.020); RANGE OVER THE 2017-PROTOCOL RUNS (4 DATASETS $\times$ 6 SIZES, $B = 50$, 5 TRIALS); EVEN-DEGREE CONTROL AND SIX-INPUT CONSTRUCTION RELATIVE TO THE NULL (MEAN OVER THOSE RUNS); FLY HASH OVER LSH ON MNIST AT $k = 4$.

hemisphere	sex	cells	inputs	fan CV	Q z	odours $\Delta\%$ (p)	shared $\Delta\%$ (p)	protocol $\Delta\%$	even	six	fly/LSH
MaleCNS R	m	1,886	5.28	0.53	+2.8	-2.1 (0.020)	-2.6 (0.020)	-10.5 to +3.1	+3.7	+5.4	2.3
MaleCNS L	m	1,865	5.30	0.54	+5.0	-1.0 (0.257)	-0.8 (0.376)	-8.0 to +11.2	+4.2	+6.0	2.2
Hemibrain R	f	1,761	6.05	0.50	-0.0	-0.6 (0.515)	-0.0 (0.990)	-6.3 to +7.4	+2.9	+2.9	2.3
FlyWire R	f	2,406	5.00	0.47	+11.8	-2.8 (0.020)	-2.7 (0.020)	-8.9 to +4.1	+1.4	+4.2	2.4
FlyWire L	f	2,378	5.12	0.49	+17.5	-1.6 (0.099)	-1.6 (0.079)	-6.6 to +3.6	+3.0	+4.2	2.9
BANC R	f	2,046	5.03	0.57	+6.0	-3.0 (0.020)	-3.0 (0.020)	-8.9 to -0.1	+3.3	+4.7	2.3
BANC L	f	2,089	5.18	0.55	+9.1	-1.3 (0.257)	-2.2 (0.059)	-9.0 to -0.2	+1.8	+4.5	2.3

TABLE IV

ODOURS, MALECNS: CONNECTOME AGAINST THE NULL ($B = 200$) AND COMPARISON MATRICES (RELATIVE TO THE NULL MEAN, %). *PRIMARY.

k	conn.	null	$\Delta\%$	p	p_{Holm}	fan flat	6 unif.	balanced
8	0.1973	0.2034 ± 0.0085	-3.0	0.468	1.000	+3.7	+26.7	+3.2
16	0.2879	0.2933 ± 0.0062	-1.8	0.438	1.000	+2.7	+9.7	+3.2
32	0.3531	0.3575 ± 0.0045	-1.2	0.338	1.000	+3.3	+3.0	+3.6
64	0.3963	0.4001 ± 0.0039	-0.9	0.348	1.000	+3.4	+0.7	+3.4
92*	0.4063	0.4148 ± 0.0035	-2.1	0.020	-	+3.2	+0.1	+3.2

and -3.0% to -0.0% on the shared glomeruli. In the full MaleCNS analysis (Fig. 4a, Table IV) the connectome scores 0.4063 against 0.4148 ± 0.0035 , -2.06% (two-sided $p = 0.020$; 0.100 with all five sizes as one family); the bootstrap 90% interval is -3.1% to -0.8%, equivalence at $\pm 5\%$ holds and at $\pm 2.5\%$ it does not hold, the interval covered the true value in 100% of 100 datasets simulated from one fitted generator, and the two-sided test has 80% power at 2.5% downward and 3.0% upward. The deficit depends on the input (Table V): imputing unmeasured responses gives -0.7% to +0.6% ($p \geq 0.439$), and all 46 glomeruli with data give no difference.

The pairing itself is structured in 6 of 7 hemispheres (Q z from -0.0 to 17.5), consistent with [5]; the hemibrain is the exception, indistinguishable from its null by Q . Structure in the pairing does not translate into better retrieval.

TABLE V

ODOURS, MALECNS, PRIMARY SIZE, ONE-FACTOR VARIATIONS ($B = 40$ NULLS EACH): RELATIVE DIFFERENCE (%) AND TWO-SIDED p .

variation	$\Delta\%$	p
Baseline	-2.2	0.049
Left hemisphere	-1.1	0.244
≥ 5 -synapse threshold	-2.3	0.098
Missing data (3 imputations)	-0.7 to +0.6	0.439–0.780
Coverage thresholds (2)	-1.7 to -1.4	0.049–0.341
All 46 glomeruli with data, imputed	-0.0	1.000
Mixture seeds (4)	-2.4 to -1.6	0.049–0.244
Mixture size 2 / 2–10	-2.0 to -0.9	0.146–0.195
Measured odorants only	-3.9	0.049
Normalised / angular metric	-2.4 to -1.7	0.049
Tie rules (index, include, 2 seeds)	-2.4 to -2.1	0.049
Synapse-weighted	-1.1	0.488
Synthetic inputs (3)	-3.1 to +0.8	0.049–0.244

D. The cost lies in glomerular fan-out

Why is random wiring slightly better? The controls change one degree sequence at a time at equal connection count (Fig. 5a). Making fan-out even, so that every glomerulus reaches the same number of Kenyon cells, raises AP on image and word input by +1% to +18%, increasingly with k ; making inputs per cell even does not (-7% to +6%). The 2017 construction behaves like the even-fan-out control. In every hemisphere the even-degree control beats the null, by +1.4% to +4.2% on average (Fig. 5b). The measured fan-out

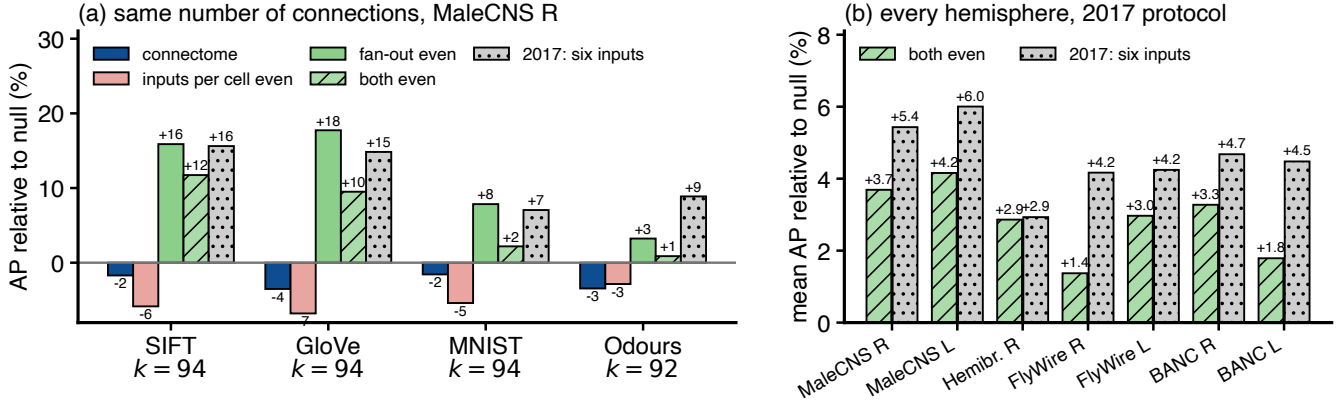


Fig. 5. Degree controls, AP relative to the null mean. (a) MaleCNS right at the largest hash size: the connectome, and matrices with the same number of connections but inputs per cell even, fan-out even, or both; the 2017 construction changes the connection count. (b) The even-degree control and the six-input construction in every hemisphere, mean over the 2017-protocol runs.

is strongly skewed (Fig. 1b; up to 473 cells for DC1 against a mean near 195), so a few glomeruli take part in many cells’ drive. On odours the same holds with smaller effects: even fan-out improves AP at every hash size (+2% to +4%), while even inputs per cell helps only at the smallest sizes. Row-centring, part of the 2017 protocol, alters winner order only when inputs per cell differ; removing it leaves image and word results unchanged but changes the odour results strongly at small k , where drive on non-negative input then tracks the number of inputs (supplement).

V. DISCUSSION

The random matrix of 2017 was a justified idealisation for similarity search. Put in its place, four measured connectomes keep the fly hash’s per-cell advantage and retrieve within a few percent of degree-preserving random matrices, slightly worse, in the direction reported for odour discrimination by [5]. Their pairing is mostly structured but does not help this task; their fan-out is skewed and costs retrieval. That skew may serve other ends, such as innate valence [5] or novelty detection [22], which we do not test.

The algorithmic claim needs a qualifier. The fly hash’s advantage is per active cell: real-valued Gaussian projections given the same projection arithmetic retrieve better on every dataset and input dimension we tested. Against one-bit codes, and at the circuit’s own scale, the comparison is closer. The finding on fan-out suggests an improvement, a fly hash with even glomerular fan-out at equal cost, which we leave to future work.

VI. LIMITATIONS

Four animals, one male; differences between connectomes mix biology and reconstruction. Our reimplementations recover the published pattern but not its absolute values, and two protocol details come from later code. Image and word input is compressed to 51 dimensions and assigned to glomeruli arbitrarily. The cross-connectome protocol runs use 50 nulls

and are effect estimates. The odour deficit depends on treating unmeasured DoOR entries as zero. The model is binary and rate-free and ignores inhibition beyond winner-take-all.

VII. CONCLUSION

The 2017 fly hashing result holds in a reimplementations and survives the substitution of four connectomes, which behave like degree-preserving random matrices and retrieve slightly worse because their glomerular fan-out is skewed. The algorithm’s advantage over LSH is per active cell, not per operation. Code, data checksums, results and a supplement are at <https://github.com/ssenge/FlyHash-Connectome>.

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